MiKA-HAD: Minority-Augmented KAN-Based Hybrid Self-Supervised Framework for Hyperspectral Anomaly Detection

Dandan Ma, Zhuozhao Liu, Zhiyu Jiang, Yi Zheng, and Qi Wang

***Abstract*—In recent years, autoencoder (AE)-based models have demonstrated significant potential in hyperspectral anomaly detection (HAD). However, the inherent class imbalance between anomaly and background samples, coupled with substantial intraclass heterogeneity within background components, often leads traditional models to excessively capture majority-class background features, thereby weakening their discriminative capabilities for minority-class backgrounds and anomaly targets. Concurrently, conventional methods fail to sufficiently model nonlinear features and adapt to complex environments, undermining their detection performance and robustness in intricate environmental interference. To address these challenges, we propose MiKA-HAD, a minority-augmented Kolmogorov–Arnold network (KAN)-based hybrid self-supervised framework for HAD. It incorporates a density-based clustering-guided sample balancing strategy that dynamically synthesizes minority background samples with spectral–spatial diversity through self-supervised learning. This approach achieves feature space rebalancing while effectively suppressing noise interference on anomaly boundaries. To overcome high-dimensional data complexity and nonlinear feature extraction challenges, we construct a hybrid self-supervised network with a KAN-based multipath collaborative module that orchestrates synergy between local sensitivity preservation, nonlinear feature modeling, and global consistency maintenance. This tripartite architecture establishes a dynamic equilibrium that enhances representation capability and environmental robustness. Extensive experiments show that MiKA-HAD achieves superior detection accuracy and stability compared to existing approaches, particularly in complex environments with varying noise conditions. The framework establishes a new paradigm for robust HAD by addressing both class imbalance bias and nonlinear feature modeling limitations.**

***Index Terms*—Anomaly detection, autoencoder (AE), hyperspectral image, Kolmogorov–Arnold network (KAN), self-supervised.**

Received 11 August 2025; revised 25 September 2025; accepted 14 November 2025. Date of publication 20 November 2025; date of current version 4 December 2025. This work was supported in part by the Young Scientists Fund of the National Natural Science Foundation of China under Grant 62301444, in part by Guangdong Basic and Applied Basic Research Foundation under Grant 2025A1515011365, in part by the Natural Science Basic Research Program of Shaanxi under Grant 2024JC-YBMS-479, and in part by the Key Research and Development Projects of Shaanxi Province under Grant 2024GX-YBXM-077. *(Dandan Ma and Zhuozhao Liu contributed equally to this work.) (Corresponding author: Zhiyu Jiang.)*

Dandan Ma is with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China, and also with Shenzhen Research Institute, Northwestern Polytechnical University, Shenzhen 518057, China.

Zhuozhao Liu is with the School of Artificial Intelligence, Optics and Electronics (iOPEN) and the School of Computer Science, Northwestern Polytechnical University, Xi'an 710072, China.

Zhiyu Jiang and Qi Wang are with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: jiangzhiyu@nwpu.edu.cn).

Yi Zheng is with the School of Computer Science, Northwestern Polytechnical University, Xi'an 710072, China.

Digital Object Identifier 10.1109/TGRS.2025.3635167

I. Introduction

HYPERSPECTRAL anomaly detection (HAD) enables the precise identification of targets by comparing pixels with distinct spatial and spectral characteristics to their adjacent background, even when prior data is unavailable [1]. This technology overcomes the reliance on target prior knowledge, demonstrating unique adaptability in unknown and complex scenarios. With the development of remote sensing platforms [2], [3], HAD has emerged as a frontier research direction in the field of multidimensional information sensing, deeply integrated into significant engineering practices such as crop stress monitoring in precision agriculture [4], [5], disaster life detection in emergency management [6], and concealed target localization in national defense security [7].

Statistically based HAD methods operate under probabilistic assumptions, where background pixels conform to predefined statistical distributions while anomalies manifest as significant deviations. The Reed–Xiaoli (RX) detector [8] identifies anomalous regions through Mahalanobis distance calculation between target pixels and background distributions under Gaussian distribution assumptions. Kwon and Nasrabadi [9] pioneered the integration of reproducing kernel Hilbert space theory to characterize high-order spectral relationships in nonlinearly separable scenarios. Zhao et al. [10] enhanced robustness through a hybrid framework combining kernelization with robust iterative estimation. Li et al. [11] extended the isolation forest to kernel space, leveraging both global statistics and local spatial–contextual information. Despite their computational efficiency, statistical methods remain constrained by simplified distribution assumptions and often fail in scenarios with complex backgrounds.

Data representation-based methods identify anomalies by computing reconstruction errors from background dictionaries, where normal pixels exhibit lower residuals compared to anomalies. Li et al. [12] introduced a joint sparse representation-based detector incorporating atomic decomposition from over-complete dictionaries. Zhao et al. [13] devised a quantitative sparsity assessment methodology improving dis-

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:55:13 UTC from IEEE Xplore. Restrictions apply.

criminative capability between the background and anomaly. Li and Du [14] introduced collaborative representation detection, further optimized using spatial contextual information and saliency analysis [15]. The low-rank decomposition extracts principal data features, while the sparse decomposition isolates deviations from normal patterns. Yang et al. [16] employed the low-rank background property to decompose HSI data into background and anomaly components. Ma et al. [17] formulated the background component as a low-rank tensor, achieving effective background-residual separation. This framework circumvents Gaussian assumptions in traditional models to some extent but faces challenges in handling noise-corrupted dictionaries and computational complexity for large-scale HSIs.

Deep learning-based methods have been established as the predominant approach for anomaly detection, primarily employing autoencoder (AE) architectures to model background distributions through reconstruction error analysis [18]. Wang et al. [19] combined fully convolutional AEs with adaptive weighted loss functions to suppress anomalous reconstruction errors. Aghili et al. [20] developed dual-domain AEs, separating anomalies from background components through nonnegative matrix factorization. Various sampling optimization strategies have been explored, including diagonal subsampling for reconstruction consistency [21], random masking [22], and multiscale masking [23] for enhanced background feature learning. For multimodal feature fusion, researchers have introduced dual-window masking [24], feature aggregation mechanisms [25], and spatial–spectral cross-guided frameworks [26]. In adversarial learning paradigms, Li et al. [27] proposed an unsupervised transfer detection model with global–local self-attention modules, while Wang et al. [28] developed a capsule differential adversarial reconstruction network. Chen et al. [29] proposed a double-window iterative spectral diffusion and inverse reconstruction method, which achieved accurate background estimation. He et al. [30] improved the Transformer network to capture deep reconstruction features from global to local views in the spatial-spectral domain. Fu et al. [31] proposed a multiscale Mamba network to enhance the representation of the background and suppress the reconstruction of anomalies. Liu et al. [32] proposed adaptive dual-domain learning based on the spatial spectrum to extract high-frequency abnormal signals in images. Although these methods have significantly improved detection performance, their reconstruction processes remain susceptible to interference from low signal-to-noise ratio data, resulting in lower detection accuracy.

In hyperspectral imagery, background samples typically dominate in quantity over anomalies while exhibiting significant intraclass heterogeneity. In hyperspectral images, background samples typically dominate in terms of quantity while exhibiting significant intraclass heterogeneity. Specifically, background samples often account for over 95% of the total sample count, and within these background samples, there are multiple distinct background types, such as trees, rivers, and shadows. The proportions of these different background samples vary greatly. Common backgrounds such as trees and rivers typically account for over 90% of the background samples, referred to as majority-class background samples, while uncommon backgrounds such as shadows and fog account for less than 10% of the samples, referred to as minority-class background samples. Conventional models tend to prioritize learning features from the majority-class background samples during training, resulting in inadequate characterization of minority-class backgrounds. This limitation consequently hampers the model's ability to discern subtle distinctions between minority-class backgrounds and anomalous targets, degrading detection performance. Moreover, traditional approaches demonstrate insufficient robustness in complex scenarios due to their limited capacity for nonlinear feature modeling and adaptability to intricate environmental interference [33]. To address these issues, we propose MiKA-HAD, a minority-augmented Kolmogorov–Arnold network (KAN)-based hybrid self-supervised framework for robust HAD. First, a density-based clustering-guided strategy separates minority-class background samples and isolated anomaly noise samples. These isolated samples are considered potential anomalies. Our hybrid self-supervised framework integrates AEs and generative adversarial networks (GANs) to implement dynamic minority-class sample augmentation, explicitly guiding the feature learning process toward these underrepresented categories. This strategy effectively enhances the network's discriminative capability in distinguishing between anomalous samples and minority-class background counterparts. Furthermore, a KAN-based multipath collaborative module achieves synergistic complementation of local sensitivity, nonlinear feature modeling, and global consistency, which enhances the nonlinear modeling capability and robustness in complex environments. The contributions of this research are outlined as follows.

1) A density-based cluster-guided sample balancing strategy is developed, where a generator network dynamically synthesizes minority background samples with spectral–spatial diversity, effectively mitigating feature learning bias induced by class imbalance. This strategy improves the detection performance through feature space rebalancing, while greatly suppressing the interference of background noise on the anomaly boundaries.
2) A KAN-based multipath collaborative module is proposed that integrates three complementary mechanisms: spline interpolation for localized feature precision, radial basis function (RBF) mapping for nonlinear representation, and linear transformations for global structural coherence. This novel joint modeling strategy effectively addresses the limitations of conventional approaches in characterizing high-dimensional nonlinear features across complex terrain regions.
3) Our MiKA-HAD framework has been validated through extensive experiments on six public datasets comprising ten real-world scenes with an overall performance gain of 2.16%, demonstrating superior detection accuracy and stability, particularly in complex environments with varying noise conditions. The experimental results confirm the effectiveness of our approach in addressing both

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:55:13 UTC from IEEE Xplore. Restrictions apply.

class imbalance bias and nonlinear feature modeling limitations.

The rest of this article is arranged as follows. The related works are briefly reviewed in Section II. Section III describes the details of the proposed method. In Section IV, the experimental results on real hyperspectral datasets are reported and analyzed. Finally, the conclusions are drawn in Section V.

II. Related Works

A. Autoencoder

AE, as an unsupervised deep neural network architecture, has gained prominence in HAD due to its distinctive feature extraction capability and noise robustness. The fundamental structure comprises symmetrical encoder–decoder components that collaboratively learn low-dimensional representations through data reconstruction [18].

In HAD applications, AE's effectiveness stems from hyperspectral imagery's inherent statistics: background pixels dominate while anomalies are extremely sparse [34]. During training, gradient descent prioritizes learning spectral mixing patterns of background regions, whereas the modeling capacity for rare anomalies remains underdeveloped. This feature learning bias creates a natural detection mechanism where anomalous pixels generate prominent reconstruction residuals compared to background components.

For a given hyperspectral pixel x, the encoder uses a trainable weight matrix W and a bias vector b to map the input sample to potential features z

$$z = f(Wx + b) \tag{1}$$

where $f(\cdot)$ is usually an element-by-element activation function such as ReLU and its variants for deeper architectures, or sigmoid for bounded outputs. The decoder then reconstructs the input data from the latent representation z using a separate weight matrix W' and bias b', followed by the same or different activation function

$$\hat{x} = f\left(W'z + b'\right). \tag{2}$$

By minimizing the reconstruction error, the network forces the latent code z to encapsulate dominant data features.

B. Generative Adversarial Network

GANs model data distributions through adversarial interplay between two modules [35]. The generator G maps latent variables z to synthetic samples $G(z)$, aiming to approximate the real data distribution, while the discriminator D evaluates sample authenticity via binary cross-entropy loss

$$\min_G \max_D \mathcal{L}_{(G,D)} = \min_G \max_D \mathbb{E}_{x\sim p_{\text{real}}(x)}\left[\log D(x)\right] + \mathbb{E}_{z\sim p_{z(z)}}\left[\log\left(1 - D(G(z))\right)\right] \tag{3}$$

where $\mathcal{L}_{(G,D)}$ is usually a binary cross-entropy loss function, x is real data, and z is a latent variable. At the Nash equilibrium, D fails to distinguish real and generated samples, indicating that G has successfully captured the essential data characteristics. Theoretically, GANs minimize Jensen–Shannon divergence for distribution alignment, yet practical improvements like Wasserstein distance and spectral normalization are often adopted to mitigate gradient vanishing or mode collapse.

Unlike conventional linear approaches, GANs establish a nonlinear characterization of HSI backgrounds through their inherent architecture, which enables robust adaptation to terrain heterogeneity and illumination fluctuations [36]. The discriminative component progressively refines its capability to identify nuanced spectral divergences between intricate backgrounds and anomalous targets through iterative adversarial optimization. Concurrently, the generative module produces diverse normal spectral samples that comprehensively represent typical surface characteristics, thereby improving the generalization capacity of detection frameworks through data-driven learning. This adversarial framework fundamentally learns background distributions directly from unlabeled hyperspectral data while maintaining environmental fidelity.

C. KAN

The Kolmogorov–Arnold representation theorem establishes that every multivariate continuous function can be expressed as a finite superposition of two-parameter univariate functions. This principle implies that complex high-dimensional relationships can be represented through simpler 1-D transformations, reducing computational complexity while preserving functional expressiveness. In particular, for any continuous function $f : [0,1]^d \to \mathbb{R}$, there exist univariate continuous functions Φ_q and $\phi_{q,p}$ such that

$$f(x_1, x_2, \ldots, x_d) = \sum_{q=1}^{2d+1} \Phi_q \left(\sum_{p=1}^{d} \phi_{q,p}\left(x_p\right) \right) \tag{4}$$

where $\Phi_q : \mathbb{R} \to \mathbb{R}$ and $\phi_{q,p} : [0,1] \to \mathbb{R}$, x_p is the input features, and d is the dimension of the input.

Based on this insight, learning a high-dimensional function essentially reduces to learning a polynomial number of 1-D functions. However, in practical scenarios, these 1-D functions may exhibit nonsmooth or even fractal characteristics, rendering them fundamentally unlearnable [37]. This critical limitation has historically hindered the application of the Kolmogorov–Arnold theorem in machine learning domains. To address this challenge, Liu et al. [38] postulated that most functions encountered in scientific and daily contexts typically demonstrate smoothness and sparse compositional structures, suggesting that network learning failures rarely occur under practical conditions. By extending neural networks to arbitrary widths and depths, they proposed the KANs

$$\text{KAN}(x) = \left(\Phi_{L-1} \circ \Phi_{L-2} \circ \cdots \circ \Phi_1 \circ \Phi_0\right) x \tag{5}$$

where l is the number of stacked layers and x is the input features.

The KAN demonstrates exceptional performance in processing high-dimensional spectral–spatial data. Leveraging their superior function approximation capabilities, the KAN can effectively extract discriminative features from hundreds of spectral bands. This property is particularly advantageous for hyperspectral data, where nonlinear spectral interdependencies align naturally with KAN's decomposition approach.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:55:13 UTC from IEEE Xplore. Restrictions apply.

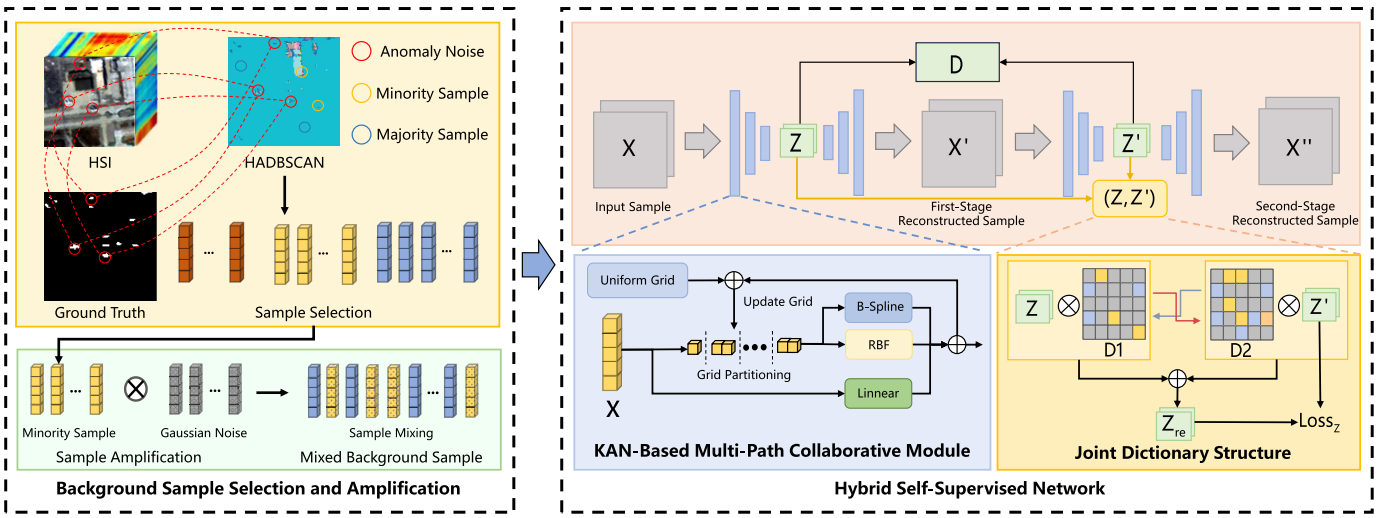


Fig. 1. Overall framework of our proposed MiKA-HAD.

Compared with a multilayer perceptron, the KAN achieves comparable or superior accuracy with fewer parameters, thereby mitigating overfitting risks under limited annotated samples. Its hierarchical decomposition aligns with the intrinsic low-dimensional structure of hyperspectral data, enabling interpretable feature learning.

III. Proposed Method

The overall framework of our proposed MiKA-HAD (minority-augmented KAN-based hybrid self-supervised framework for HAD) is illustrated in Fig. 1. First, a background sample selection and amplification strategy is developed to dynamically identify and augment minority-class background samples, effectively rebalancing the feature space. Second, a hybrid self-supervised network is constructed to enhance the network's ability to discriminate minority class samples and anomalous targets by dynamically generating diverse minority-class background samples. Within this hybrid network, we further design a KAN-based multipath collaborative module that orchestrates synergy among local sensitivity preservation, nonlinear feature modeling, and global consistency maintenance for augmenting the model's adaptability and flexibility to complex hyperspectral scenes.

A. Background Sample Selection and Amplification

In HAD, background pixels typically constitute 95%–99.9% of the total imagery, while anomalous targets exist in sparse distributions [39]. Meanwhile, the background regions themselves exhibit significant heterogeneity. As illustrated in Fig. 2 using the Los Angeles-II dataset, while background pixels account for 99.13% of the total, they exhibit a pronounced internal imbalance: 93.65% represent majority-class backgrounds (vegetation and soil), while only 5.48% constitute minority-class backgrounds (shadows and moisture changes)—a proportion barely exceeding that of anomalous samples (0.87%). This imbalance creates a critical detection bottleneck, as models struggle to distinguish subtle spectral differences between minority-class background samples and genuine anomalies, which affects the detection efficiency in complex scenarios.

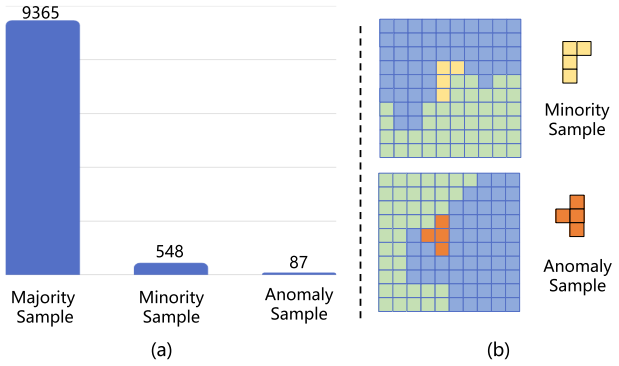


Fig. 2. (a) Clustering results on the Los Angeles-II dataset. The majority-class background samples account for more than 93% of the total. (b) Minority-class background may be similar in appearance to anomalies, where green and blue pixels represent the majority-class background, yellow pixels represent the minority-class background, and orange pixels represent anomalies.

This imbalance fundamentally alters model optimization trajectories during training. Specifically, during stochastic gradient descent, parameter updates are governed by

$$\nabla_\theta \mathcal{L}(\theta) = \frac{1}{\mathcal{N}} \sum_{x \in \mathcal{N}} \frac{\partial \mathcal{L}(x;\theta)}{\partial \theta} = \frac{\mathcal{N}_{\text{maj}}}{\mathcal{N}} \cdot \nabla \mathcal{L}_{\text{maj}} + \frac{\mathcal{N}_{\text{min}}}{\mathcal{N}} \cdot \nabla \mathcal{L}_{\text{min}} + \frac{\mathcal{N}_a}{\mathcal{N}} \cdot \nabla \mathcal{L}_a \quad (6)$$

where $\mathcal{N}$ denotes the total number of samples processed. $\mathcal{N}_{\text{maj}}$, $\mathcal{N}_{\text{min}}$, and $\mathcal{N}_a$ represent the number of majority-class background samples, minority-class background samples, and anomalous samples, respectively. $(\partial \mathcal{L}(x;\theta))/(\partial\theta)$ represents the gradient contribution of a single input sample x with respect to parameters θ. $\nabla \mathcal{L}_{\text{maj}}$, $\nabla \mathcal{L}_{\text{min}}$, and $\nabla \mathcal{L}_a$ denote the average loss gradients computed over these three categories, respectively. The terms $(\mathcal{N}_{\text{maj}})/\mathcal{N}$, $(\mathcal{N}_{\text{min}})/\mathcal{N}$, and $(\mathcal{N}_a)/\mathcal{N}$ represent their proportional weights within the dataset, satisfying $\mathcal{N}_{\text{maj}} + \mathcal{N}_{\text{min}} + \mathcal{N}_a = \mathcal{N}$. Given that

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:55:13 UTC from IEEE Xplore. Restrictions apply.

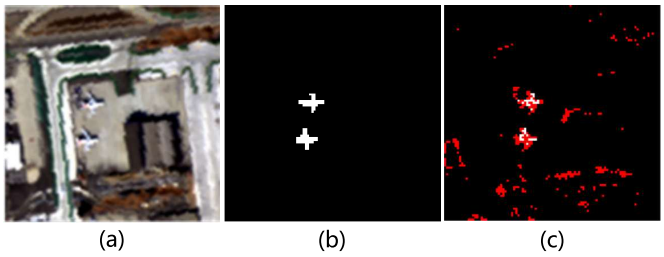


Fig. 3. (a) Pseudo-color image of the Los Angeles-II dataset. (b) Ground-truth annotation of the Los Angeles-II dataset, where white pixels denote anomalous regions. (c) Distribution of isolated pixels identified by HDBSCAN clustering: the red pixels represent these isolated pixels, which constitute merely 3.10% of the total pixels yet cover 60.92% of the actual anomalous areas.

$N_{\text{maj}} \gg N_{\text{min}} \approx N_a$, the resultant gradient predominantly optimizes for the majority-class background. This progressively shifts the decision boundaries toward the feature spaces of the majority class, degrading the model's capacity to distinguish minority backgrounds from anomalies.

To address this challenge, MiKA-HAD incorporates a density-based clustering-guided background sample selection and amplification strategy. After PCA-based dimensionality reduction and normalization, we leverage the hierarchical density-based spatial clustering of applications with noise (HDBSCAN) method [40] to identify and separate minority-class background samples from majority classes and potential anomalies. HDBSCAN combines the ideas of density clustering and hierarchical clustering, using a density hierarchy tree to represent the clustering structure of the data. This approach exploits the inherent spectral–spatial distribution patterns in hyperspectral data: majority backgrounds typically form dense, continuous clusters with strong spatial correlation, while minority backgrounds and anomalies manifest as sparse, isolated structures due to their spectral deviation from dominant patterns. With hyperparameters min-cluster size mc and min-samples ms, the HDBSCAN algorithm can generate multiple sample clusters and isolated noise points. We designate clusters with fewer than k elements as sparse clusters (minority-class backgrounds), where the threshold k is determined dynamically to ensure sparse clusters constitute $r\%$ of the total clustered pixels. This adaptive thresholding approach ensures consistent minority sample identification across varied datasets. Importantly, we exclude isolated noise points (potential anomalies) to prevent anomaly contamination during background modeling. Fig. 3 illustrates the HDBSCAN clustering results on the Los Angeles-II dataset. In Fig. 3(b) and (c), black pixels represent background regions, white pixels denote true anomaly areas, and red pixels mark isolated pixels identified by HDBSCAN. These isolated pixels account for only 3.10% of the total pixels but cover 60.92% of the anomaly regions. To improve anomaly suppression, we exclude the isolated pixels from network training. Notably, erroneously excluded background regions can be compensated through data augmentation from neighboring pixels, whereas true anomalies are difficult to recover due to spectral differences. This further widens the proportion gap between minority-class backgrounds and anomaly samples.

After identification, we amplify minority-class samples through a controlled augmentation strategy. The sparse cluster samples are expanded m-fold, where m is a hyperparameter controlling the augmentation multiple. To enhance feature space diversity while preserving spectral characteristics, we introduce Gaussian noise following $\mathcal{N}(0, 1 \times 10^{-7})$ to each augmented sample. This choice leverages the natural property of Gaussian noise to approximate multiple independent perturbation sources without directional bias [41], effectively expanding the feature space representation of minority-class samples while maintaining their intrinsic spectral signatures. This balanced augmentation approach enables the model to develop more discriminative representations for both majority and minority background classes, significantly enhancing its ability to distinguish between anomalies and complex background variations.

B. Hybrid Self-Supervised Network

We propose a hybrid self-supervised network combining AEs and GANs to enhance the ability of HAD networks to characterize minority background samples through a multistage feature optimization mechanism. The network employs a cascaded structure with two AEs and a GAN. After initial feature compression and reconstruction by the first-stage AE, the network performs adversarial optimization via the GAN, followed by final reconstruction through the second-stage AE, forming a self-supervised closed loop. This hybrid architecture offers dual advantages: it prevents pattern collapse in GAN training through the cascaded reconstruction constraints of the AEs, while leveraging the adversarial mechanism to focus model learning on subtle spectral differences between minority-class backgrounds and anomalies, addressing a key limitation of pure AE approaches in sparse sample learning. Additionally, we introduce a joint dictionary structure that ensures semantic consistency during feature refinement through self-consistent constraints. The network comprises three complementary components: a feature encoding module, an adversarial generation module, and a feature refinement module.

In the encoding phase, the first-stage autoencoder Ae, consisting of an encoder En and a decoder De, compresses and reconstructs the input pixel x to generate the potential feature z and the reconstructed sample x'. In the generation phase, the generator Ge, which has the same structure as the En, extracts the refinement feature z' from x'. Meanwhile, the discriminator Di distinguishes between the original feature z and the refined feature z' through adversarial training. During the refinement stage, the second-stage autoencoder Ae$'$, consisting of a generator Ge and a decoder De$'$, reconstructs z' into x'', establishing a cascading optimization path with the first-stage AE. To enhance the network's capacity for modeling complex high-dimensional data and extracting nonlinear features, we replace traditional linear layers in the AEs with our KAN-based multipath collaborative module, detailed in Section III-C.

Notably, a joint dictionary structure is introduced to map potential spatial features, ensuring $L2$ distance minimization between dictionary-reconstructed features z_{re} and z'. This approach maintains distributional consistency between

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:55:13 UTC from IEEE Xplore. Restrictions apply.

the original z and refined z' while preserving background purity through sparse dictionary constraints. The joint dictionary learning framework captures multiview background representations using two co-trained dictionaries. The input features z and z' undergo normalization before initialization of orthogonal random dictionary matrices D_1 and D_2 and the corresponding sparse coefficient matrices X_1 and X_2. During the iterative optimization phase, we alternately perform three core steps: sparse coding, dictionary updating, and cross-dictionary regularization. Sparse coding is achieved by solving for the coefficient matrices using least-squares with ReLU-enforced nonnegativity. Dictionary updating employs an atom strategy to compute residuals and optimize the atom vectors. Cross-dictionary regularization enhances information sharing between the two dictionaries. The structure's objective function is as follows:

$$\begin{aligned} D1 &= (1-\lambda)\, D_1 + \lambda D_2 \\ D2 &= (1-\lambda)\, D_2 + \lambda D_1 \\ \text{Loss}_z &= \left\| z' - \frac{1}{2}\left(D_1 X_1 + D_2 X_2\right) \right\|_F^2 \end{aligned} \tag{7}$$

where $\|\cdot\|_F^2$ represents feature reconstruction error and λ controls feature sharing strength between dictionaries.

The principal AE Ae loss incorporates the pixel-level reconstruction error, feature alignment error, and adversarial discrimination loss

$$\begin{aligned} \text{Loss}_{\text{Ae}} = &\sum_{i=1}^{N} \left(x_i - {x'}_i\right)^2 + \text{Loss}_z \\ &- \mathbb{E}_{z\sim p_{\text{real}(z)}}\left[D\left(z\right)\right] \end{aligned} \tag{8}$$

where x_i is the input sample, ${x'}_i$ denotes the first-stage reconstructed sample, and $\mathbb{E}_{z\sim p_{\text{real}(z)}}[D(z)]$ corresponds to the loss where the discriminator Di classifies the real sample z as real. Similarly, AE Ae$'$ loss is

$$\begin{aligned} \text{Loss}_{\text{Ae}'} = &\sum_{i=1}^{N} \left({x'}_i - {x''}_i\right)^2 + \text{Loss}_z \\ &- \mathbb{E}_{z'\sim p_{\text{fake}(z')}}\left[D\left(z'\right)\right] \end{aligned} \tag{9}$$

where ${x''}_i$ is the second-stage reconstructed sample and $\mathbb{E}_{z'\sim p_{\text{fake}(z')}}[D(z')]$ corresponds to the loss where the discriminator Di classifies the fake sample z' as fake. The discriminator Di loss is

$$\begin{aligned} \text{Loss}_{\text{Di}} = &-\mathbb{E}_{z\sim p_{\text{real}(z)}}\left[D\left(z\right)\right] \\ &+ \mathbb{E}_{z'\sim p_{\text{fake}(z')}}\left[D\left(z'\right)\right] + \eta\cdot \text{GP} \end{aligned} \tag{10}$$

where $\eta\cdot\text{GP}$ represents the gradient penalty term, with η controlling penalty strength.

This architecture achieves progressive feature purification through cascaded dual AEs, gradually eliminating noise interference through staged reconstruction. The adversarial training mechanism encourages generator Ge to learn more discriminative feature distributions while dynamically adjusting feature space separability through discriminator Di feedback. Moreover, the joint dictionary structure establishes self-consistent constraints between potential features, ensuring semantic consistency throughout the feature refinement process via closed-loop optimization.

C. KAN-Based Multipath Collaborative Module

Within the hybrid self-supervised network, we design a KAN-based multipath collaborative module to address the inherent complexity of hyperspectral data characterized by high dimensionality and nonlinear spectral–spatial correlations. This module replaces conventional linear layers in the AEs, substantially enhancing network flexibility and adaptability. Our design integrates three complementary mechanisms: spline interpolation for localized feature precision, RBF mapping for nonlinear feature completion [42], and linear transformations for global structural coherence. This multipath architecture establishes dynamic collaboration among specialized pathways, enabling joint optimization of local sensitivity and global consistency.

The module simultaneously activates three independent computational paths to process input data after initialization. The basic linear transformation path constructs global feature baselines through parameterized weight matrices and SiLU nonlinear activation. The B-spline basis function path performs local adaptive partitioning of input space using differentiable spline interpolation. By imposing continuity constraints on piecewise polynomial functions at preset grid nodes, it constructs a flexible feature mapping for locally complex high-dimensional data

$$\begin{aligned} B_{i,k}\left(x\right) = &\frac{x - g_i}{g_{i+k} - g_i} B_{i,k-1}\left(x\right) \\ &+ \frac{g_{i+k+1} - x}{g_{i+k+1} - g_{i+1}} B_{i+1,k-1}\left(x\right) \end{aligned} \tag{11}$$

where $B_{i,k}(x)$ is the ith basis function of order k, g_i, g_{i+k}, and g_{i+k+1} denote grid nodes, and $B_{i,k-1}(x)$ is the basis function of order $k-1$. Throughout training, the architecture adaptively optimizes the learnable weights governing dynamic and uniform grids to enhance adaptability across diverse input samples.

The RBF path uses kernel function mapping to project input features into implicit high-dimensional spaces. In contrast to conventional SiLU, RBF leverages explicit spatial distance measurements between data points and predefined reference points to establish a localized response mechanism. This architecture provides innate sensitivity to local spectral anomalies while offering physically interpretable feature representations

$$\text{RBF}\left(x\right) = \left(1 + \frac{\|x-g\|}{\sigma^2}\right)^d \tag{12}$$

where x represents the input feature, g is the grid nodes, σ is the scaling factor that governs the width of the RBF, and d is the exponent that determines the RBF shape. The output tensors from the three paths undergo scale normalization and are then summed with weighted coefficients to form the final feature representation. These weights are dynamically optimized through learnable parameters.

This multipath collaborative module offers significant complementary advantages. The spline path achieves accurate local

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:55:13 UTC from IEEE Xplore. Restrictions apply.

feature capture through linear combinations of piecewise polynomial basis functions, with error limits controlled by mesh density and spline order, providing flexibility and adaptability to locally complex features. The radial basis path relies on kernel method theory, ensuring feature space completeness through Mercer's theorem [43] while excelling at capturing nonlinear feature associations. The basic linear term maintains the basic global feature construction. This collaborative structure achieves dynamic equilibrium between local feature sensitivity and global structural coherence, enabling robust adaptation to complex scenes while preserving topological consistency across varying spatial scales.

D. Anomaly Detection

The residual image x_{res} is generated through pixel-level discrepancy computation between the original HSI and its reconstruction. An initial anomaly score map is then produced by computing the Mahalanobis distance for each residual pixel. To enhance detection sensitivity, the anomaly scores of previously excluded isolated noise pixels are strategically amplified by a factor of two. This targeted score adjustment effectively distinguishes genuine anomalies from background interference while maintaining computational efficiency, yielding a refined anomaly distribution map that optimizes the balance between detection accuracy and false alarm suppression. The anomaly score is as follows:

$$\mathrm{S}_i = \begin{cases} 2 \cdot \left(x_{\text{res}}^{(i)} - \mu\right)^{\top} \Sigma^{-1} \left(x_{\text{res}}^{(i)} - \mu\right), & i \in \Omega \\ \left(x_{\text{res}}^{(i)} - \mu\right)^{\top} \Sigma^{-1} \left(x_{\text{res}}^{(i)} - \mu\right), & \text{other} \end{cases} \tag{13}$$

where $x_{\text{res}}^{(i)}$ are residual image pixels, Σ^{-1} is the inverse covariance matrix, μ is the mean vector, and Ω is the sequence of previously excluded isolated noise pixels.

IV. Experimental Results and Analysis

This section presents a comprehensive evaluation of our proposed MiKA-HAD framework through four systematic investigations. All experimental data are from our own experiments. First, comparative experiments are conducted across several publicly available datasets with diverse backgrounds and anomaly characteristics. Subsequently, parametric sensitivity analysis examines the robustness of key hyperparameters. Then, rigorous ablation studies are implemented. Finally, anti-interference experiments verify the framework's robustness against varying noise levels. The experiment is implemented using Python 3.9 and MATLAB R2018b on an NVIDIA GeForce RTX 4090 GPU. The experiment used the Adam optimizer with 20 epochs, the AE learning rates of 0.001, the GAN learning rate of 0.0001, the dictionary collaboration degree of 0.5, and noise added to the augmented pixels as random values obeying $N(0, 1e-7)$. The dimension settings for each layer of the GAN and the two AE groups were the same, namely 40 and 80. The number of grid divisions for the B-spline is 5, and the degree of the spline is 3. The number of grid divisions for the RBF is 5, the degree is 2, and the scaling factor is 2. The PCA dimension was preset to 120, but due to the insufficient number of bands in the U-HAD dataset, the PCA dimension was set to 50 for U-HAD. All experiments were conducted with ten repeated trials to ensure reliability, with reported results representing average outcomes. The experimental analysis incorporated four principal indicators for evaluating anomaly detection performance: receiver operating characteristic (ROC) curves, their corresponding area under the curve (AUC) values, and the signal-to-noise probability ratio (SNPR). ROC provides a threshold-independent performance analysis by plotting the relationship between the probability of detection (PD) and the false alarm rate (FAR). Higher PD with a lower FAR indicates superior performance. AUC quantifies the ROC's overall discriminative power. Larger AUC values signify better detection capability, particularly useful when ROC curves intersect. The SNPR measures detector confidence in identifying genuine targets amidst interference by comparing the probability of correct detection to the probability of false alarms. A higher SNPR indicates superior separation between signal and noise distributions, implying greater reliability in discerning targets from clutter. This metric emphasizes inherent probabilistic separability over absolute energy thresholds, complementing tools like ROC analysis, which rely on threshold-dependent outcomes. Furthermore, to assess computational efficiency, the running times of all methods were recorded across all datasets, with centisecond precision.

A. Datasets Descriptions

In our comparative experiments, we utilized six publicly available hyperspectral datasets comprising ten real-world scenes. These datasets exhibit anomalies that vary in both type and size, with diverse background compositions and complex minority-class distributions, providing a comprehensive evaluation platform for assessing the anomaly detection performance.

1) Cat Island: This real dataset was collected in Cat Island by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor. The dataset contains 150×150 pixels with 188 bands, and its spatial resolution is 17.2 m. The rocks are considered as anomalies, with 19 anomalous pixels accounting for 0.08% of the total.

2) Los Angeles: This real dataset was collected in Los Angeles by AVIRIS sensors. The dataset contains the Log Angeles-I–IV subsets with 100×100 pixels, 205 bands, and a spatial resolution of 7.1 m. Aircraft and industrial facilities are considered as anomalies, with 144, 87, 170, and 232 anomalous pixels, or 1.44%, 0.87%, 1.7%, and 2.32% of the total number of anomalies, respectively.

3) Gulfport: This real dataset was collected in Gulfport by the AVIRIS sensor. The dataset contains 100×100 pixels with 191 bands, and its spatial resolution is 3.4 m. The aircraft are considered as anomalies, with 60 anomalous pixels accounting for 0.6% of the total.

4) Gainesville: This real dataset was collected in Gainesville by the AVIRIS sensor. The dataset contains 100×100 pixels with 191 bands, and its spatial resolution is 3.5 m. The ships and facilities are considered as anomalies, with 52 anomalous pixels accounting for 0.52% of the total.

5) Bay Champagne: This real dataset was collected in Bay Champagne by the AVIRIS sensor. The dataset contains $100 \times$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:55:13 UTC from IEEE Xplore. Restrictions apply.

TABLE I

AUC SCORES AND EXECUTION TIMES FOR DIFFERENT METHODS ON ALL DATASETS. THE BEST PERFORMERS PER ROW ARE BOLDED; SECOND-BEST PERFORMERS ARE UNDERLINED

Dataset	WSCF [44]	RX [8]	LRX [45]	CRD [14]	KIFD [11]	CTAD [46]	DFAN [25]	BlockNet [47]	GT-HAD [48]	MiKA-HAD
Cat Island	0.9799	0.9698	0.9901	0.9818	0.9753	0.9335	0.9889	0.9855	0.9845	**0.9940**
Time (s)	0.12	1.63	1.58	40.89	41.54	73.09	44.50	169.05	100.43	191.23
Los Angeles-I	0.8511	0.8062	0.9146	0.9588	0.9113	0.7856	0.9144	0.9173	0.8568	**0.9593**
Time (s)	0.06	0.68	0.69	33.43	33.49	36.46	12.10	167.68	40.05	216.53
Los Angeles-II	0.8802	0.8399	0.8968	0.8278	0.9739	0.8445	0.9368	0.9453	0.8701	**0.9768**
Time (s)	0.06	0.64	0.68	33.96	34.14	36.81	10.32	165.42	38.73	160.53
Los Angeles-III	0.9295	0.9272	0.9205	0.9443	**0.9560**	0.8999	0.9412	0.9295	0.8658	0.9506
Time (s)	0.06	0.65	0.68	31.58	32.47	35.64	11.05	165.00	38.99	166.72
Los Angeles-IV	0.9642	0.9702	0.9387	0.9526	0.9775	0.9606	0.9501	0.8858	0.9275	**0.9843**
Time (s)	0.06	0.65	0.68	32.87	32.96	36.06	12.27	162.27	39.64	159.59
Gulfport	0.9637	0.9525	0.7861	0.7573	0.9898	0.9176	0.9958	0.9831	0.9340	**0.9963**
Time (s)	0.06	3.13	0.65	34.25	34.56	35.62	9.39	197.76	59.57	220.17
Gainesville	0.9412	0.9498	0.9425	0.9575	0.9514	0.9089	0.9696	0.9804	0.9393	**0.9907**
Time (s)	0.06	0.73	0.72	32.57	32.67	33.77	12.68	167.38	39.70	136.97
Bay Champagne	0.9992	0.9998	0.9997	0.9996	0.9869	0.9620	0.9998	0.9992	0.9814	**0.9999**
Time (s)	0.06	0.73	0.63	33.06	33.23	33.41	10.64	168.98	39.01	146.12
UHAD-U-I	0.9032	0.9236	0.8391	0.9406	0.9304	0.5543	0.8697	0.8921	0.8982	**0.9795**
Time (s)	0.14	4.00	2.66	77.29	77.07	94.62	99.50	169.92	272.90	214.02
UHAD-U-II	0.9651	0.9759	0.9438	0.9681	0.9589	0.8232	0.9526	0.9618	0.9902	**0.9948**
Time (s)	0.14	4.00	2.64	76.87	78.29	98.05	96.50	165.89	272.39	225.59
Average	0.9377	0.9314	0.9171	0.9288	0.9611	0.8590	0.9518	0.9480	0.9247	**0.9826**
Time (s)	0.08	1.68	1.16	42.68	43.04	51.35	31.90	169.94	94.14	183.75

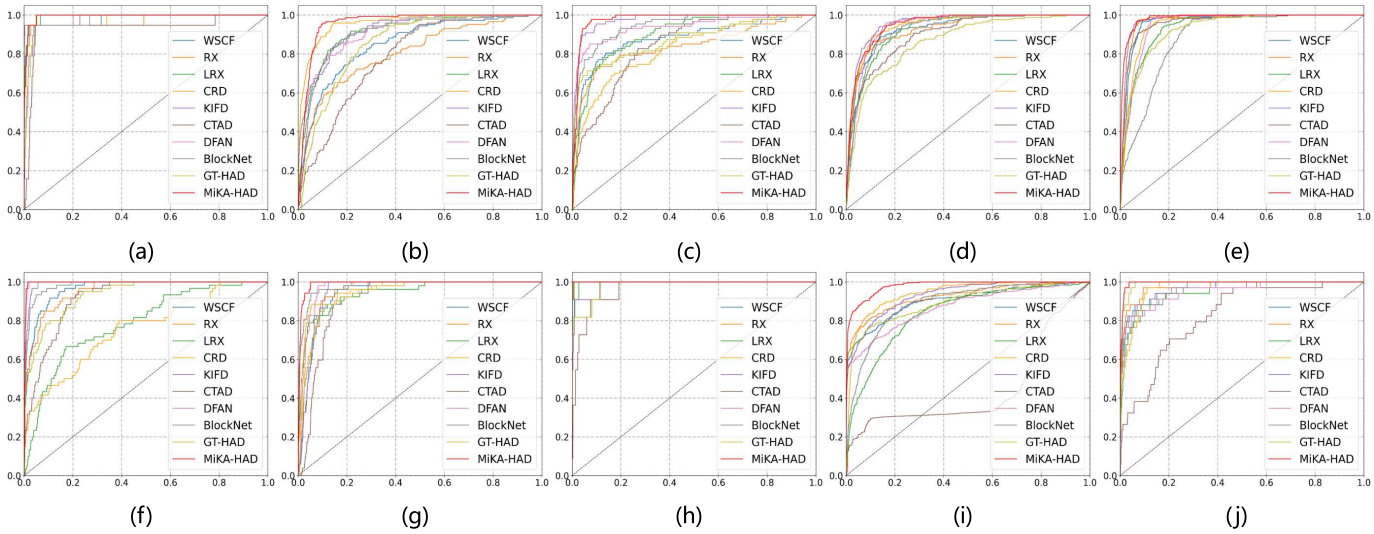


Fig. 4. ROC curves for different methods on all datasets. (a) Cat Island. (b) Los Angeles-I. (c) Los Angeles-II. (d) Los Angeles-III. (e) Los Angeles-IV. (f) Gulfport. (g) Gainesville. (h) Bay Champagne. (i) UHAD-U-I. (j) UHAD-U-II.

100 pixels with 188 bands, and its spatial resolution is 4.4 m. The rocks are considered as anomalies, with 11 anomalous pixels accounting for 0.11% of the total.

6) UHAD-U: This real dataset was collected in Guangzhou by the HyperspecI sensor. The dataset contains UHAD-U-I–II subsets, 250×250 pixels, 61 bands, and a spatial resolution of 0.053 m. Vehicles and pedestrians are considered as anomalies, with 302 anomalous pixels representing 0.48% of the total and 34 anomalous pixels representing 0.056% of the total, respectively.

B. Comparison With Other Competitors

Our MiKA-HAD framework was benchmarked against nine representative methods spanning three methodological categories to establish a comprehensive performance comparison. As for statistical-based methods, RX [8] detects anomalies using the Gaussian distribution and covariance matrices. LRX [45] improves RX by focusing on local regions for small-scale anomalies. CTAD [46] employs a checkerboard topology framework for direct high-dimensional feature mining. KIFD [11] maps hyperspectral data to kernel space for iterative refinement of anomaly maps. As for representation-based methods, WSCF [44] enhances anomaly detection through a sparse coding framework for pixel combination. CRD [14] uses collaborative representation with spatial neighborhood data for anomaly detection. As for deep learning-based methods, DFAN [25] integrates spectral attention and background statistics for effective anomaly capture. BockNet [47] is a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:55:13 UTC from IEEE Xplore. Restrictions apply.

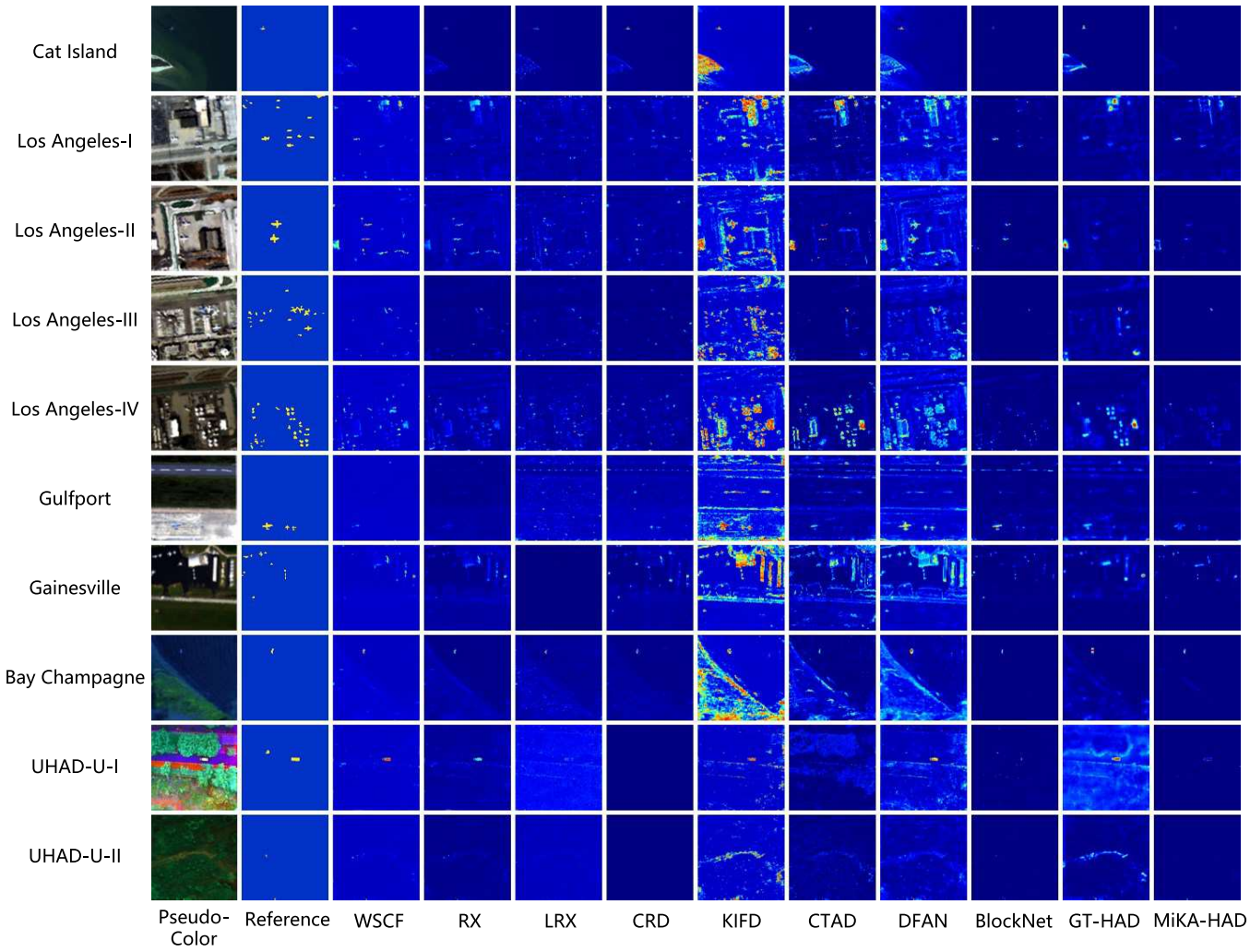


Fig. 5. Anomaly detection results of different methods on all datasets.

self-supervised network with blind spots for protected window reconstruction. GT-HAD [48] utilizes a gated Transformer network to model background and anomaly features based on content similarity.

Table I provides quantitative AUC scores and execution times metrics with the best values in bold, and Fig. 4 compares the ROC curves. The experimental results demonstrate that MiKA-HAD demonstrates exceptional detection performance across nine of the ten test scenes, with an average AUC improvement of 2.16% over the second-best method. This consistent performance advantage is particularly noteworthy in complex scenes where minority background classes are present. Furthermore, our method maintains low FARs across all datasets, achieving an optimal balance between detection sensitivity and false alarm suppression—a critical requirement for practical applications. Present investigations center on algorithmic efficacy, while forthcoming research will emphasize enhancing computational efficiency.

Fig. 5 presents the anomaly detection visualization results across all datasets. For complex scenes like the Gainesville dataset with spectral variations and low target–background contrast, traditional statistical methods, RX, LRX, and collaborative methods, CRD exhibit limited discrimination capability, manifesting as background interference, incomplete target detection, or blurred anomaly boundaries. Kernel-based approaches, KIFD, suffer from excessive false alarms due to inadequate high-dimensional space modeling, while deep learning methods, DFAN, BlockNet, and GT-HAD, show constrained scene adaptability, leading to residual background artifacts or partial target omissions. In contrast, our method achieves comprehensive anomaly localization while maintaining background purity. On UHAD-U-I datasets characterized by nonlinear mixing and low target–background contrast, whereas conventional approaches struggle to reconcile background suppression with anomaly fidelity, our method strikes an optimal balance between background suppression and accurate anomaly identification in such challenging scenarios, owing to the synergistic effect of minority sample augmentation and KAN-based feature learning.

Table II presents the SNPR scores across all datasets. The results indicate that MiKA-HAD achieves the highest SNPR values on the Bay Champagne, Gainesville, Los Angeles-III–IV, and UHAD-U-I–II datasets, ranks second on the remaining datasets, and attains the highest average SNPR

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:55:13 UTC from IEEE Xplore. Restrictions apply.

TABLE II

SNPR FOR DIFFERENT METHODS ON ALL DATASETS. THE BEST PERFORMERS PER ROW ARE BOLDED; SECOND-BEST PERFORMERS ARE UNDERLINED

Dataset	WSCF [44]	RX [8]	LRX [45]	CRD [14]	KIFD [11]	CTAD [46]	DFAN [25]	BlockNet [47]	GT-HAD [48]	MiKA-HAD
Bay Champagne	44.9929	20.4614	13.0393	23.2469	3.1485	6.617	7.442	433.7165	28.3565	**564.7276**
Cat Island	121.5729	46.2103	30.2985	40.6402	14.301	14.9404	15.6119	**598.8129**	38.2855	158.0811
Gainesville	3.4068	3.0006	0.4968	6.2016	5.5692	3.0568	3.7767	**48.3297**	6.7989	44.4419
Gulfport	6.0712	2.9604	1.6492	2.2747	6.5652	4.493	6.3791	23.9449	6.0094	**26.4955**
Los Angeles-I	4.6693	2.3016	2.8158	3.5992	4.8235	3.3216	2.9543	**24.3451**	2.809	7.0577
Los Angeles-II	9.4974	3.6243	2.9281	3.2103	5.6038	7.1192	4.2672	**39.2975**	5.4892	20.9825
Los Angeles-III	11.5749	4.7204	3.5169	5.6362	5.9628	21.3198	3.3025	28.1511	4.9186	**79.5963**
Los Angeles-IV	7.3	3.5628	3.8688	3.4137	7.2511	12.4581	3.1864	7.9446	6.6124	**18.0597**
UHAD-U-I	33.1792	7.6379	1.4991	0.8611	15.0557	0.6661	4.7556	19.8059	3.1357	**188.831**
UHAD-U-II	72.1802	16.4662	1.8803	0.862	13.2126	4.8915	8.0907	72.5102	14.9344	**280.0329**
Average	31.44448	11.09459	6.19928	8.99459	8.14934	7.88835	5.97664	129.68584	11.73496	**138.83062**

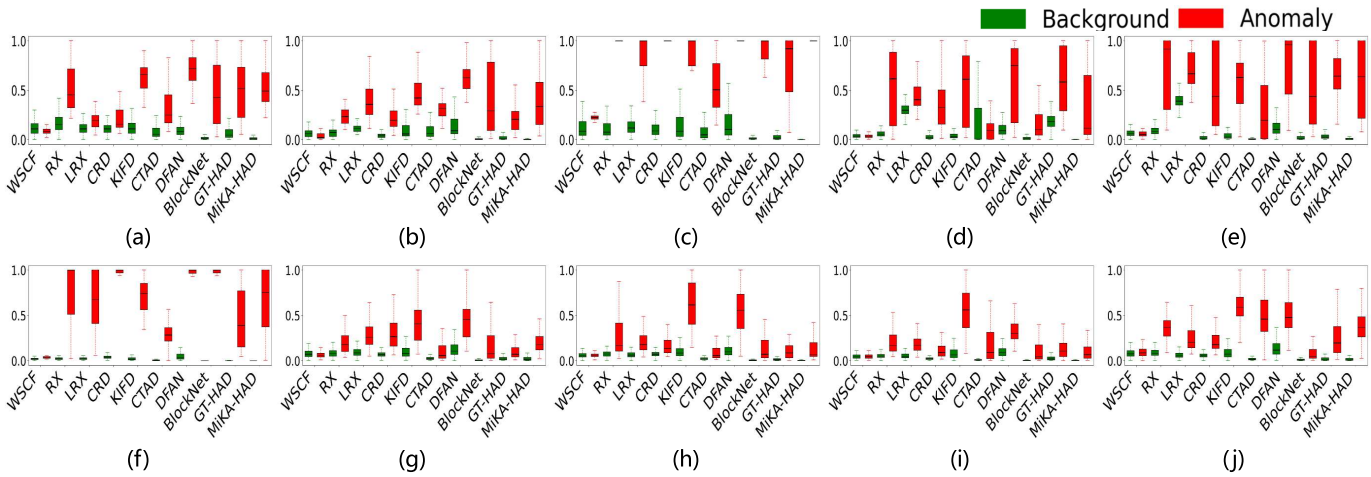


Fig. 6. Box plots of different methods on all datasets. (a) Cat Island. (b) Los Angeles-I. (c) Los Angeles-II. (d) Los Angeles-III. (e) Los Angeles-IV. (f) Gulfport. (g) Gainesville. (h) Bay Champagne. (i) UHAD-U-I. (j) UHAD-U-II.

value. This demonstrates that our method effectively detects anomalies while suppressing background noise. In contrast, methods such as KIFD and DFAN, which previously excelled in AUC experiments, did not maintain their advantage in the SNPR evaluation, showing a significant decline in ranking. This highlights the challenge they face in balancing detection probability and FAR. These results validate that our designed background sample selection and amplification module effectively differentiates anomaly samples from minority background samples, while the multipath collaborative module enhances the network's local sensitivity and nonlinear feature extraction capabilities, enabling the model to adapt to diverse complex scenarios and ultimately achieve a favorable balance between detection performance and background suppression.

Fig. 6 presents the box plot comparison of anomaly detection performance across datasets using different methods. The red boxes indicate anomaly score distributions, while the green boxes represent background scores. The spacing between boxes reflects anomaly–background separability, and the spread of green boxes directly visualizes background suppression effectiveness. Experimental results reveal that MiKA-HAD consistently achieves the optimal anomaly–background separation across most datasets, indicated by the maximal distance between red and green boxes. Furthermore, our method demonstrates superior background suppression capabilities, evidenced by the compact distribution of green boxes that indicates precise, consistent background modeling. In contrast, comparative methods exhibit varying degrees of distribution overlap between anomaly and background classes, alongside significantly wider background score distributions, indicating their susceptibility to minority-class misclassifications. This quantitative analysis confirms that our approach effectively addresses both feature learning bias from class imbalance and nonlinear modeling limitations in complex environments.

C. Ablation Experiment

This section evaluates three core components through ablation studies: background sample selection and amplification, a hybrid self-supervised network, and a multipath collaborative module. The ablation study consists of five progressive configurations: $C1$ uses only the AE, $C2$ adds background sample selection and amplification to $C1$, $C3$ integrates the hybrid self-supervised network into $C2$, $C4$ builds upon $C3$ by incorporating the KAN module, and $C5$ incorporates the KAN-based multipath collaborative module based on $C3$. As shown in Table III and Fig. 7, the comparison between $C2$ and $C1$ demonstrates that traditional models based on AE and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:55:13 UTC from IEEE Xplore. Restrictions apply.

TABLE III
AUC OF DIFFERENT COMPONENTS ON REPRESENTATIVE DATASETS

Dataset	Cases	AE	Sample Selection and Amplification	Hybrid Self-Supervised Network	KAN	Multi-Path Collaborative Module	AUC
Gulfport	C1	✓	-	-	-	-	0.9869
	C2	✓	✓	-	-	-	0.9931
	C3	✓	✓	✓	-	-	0.9941
	C4	✓	✓	✓	✓	-	0.9947
	C5	✓	✓	✓	-	✓	**0.9963**
Log Angeles-II	C1	✓	-	-	-	-	0.9542
	C2	✓	✓	-	-	-	0.9545
	C3	✓	✓	✓	-	-	0.9612
	C4	✓	✓	✓	✓	-	0.9693
	C5	✓	✓	✓	-	✓	**0.9768**

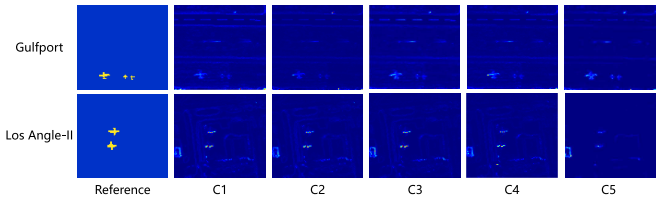


Fig. 7. Ablation visualization results on representative datasets.

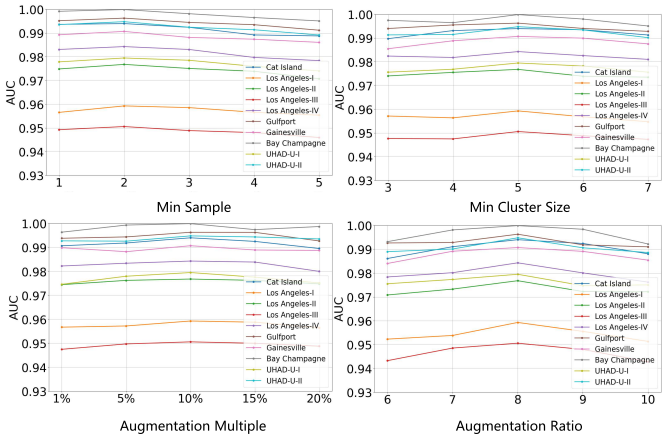


Fig. 8. AUC for different parameters on all datasets.

similar architectures indeed tend to prioritize learning features of majority-class background samples, resulting in insufficient representation of minority-class backgrounds. The sample balancing strategy in $C2$ substantially reduces false detections by suppressing background noise and enhancing minority-class representation. The evidence is particularly compelling on the Gulfport dataset, where AUC improves from 0.9869 to 0.9931, confirming our method's efficacy in addressing the class imbalance challenge. The hybrid self-supervised network in $C3$ further refines anomaly boundary definition and enhances detection uniformity through adversarial feature fusion, with AUC reaching 0.9941 on Gulfport and 0.9612 on Los Angeles-II. Meanwhile, background sample selection and amplification introduced in $C2$ effectively address the issue of sample space imbalance that traditional GANs often encounter when generating samples. Furthermore, the KAN module was introduced based on $C3$. Although false alarms still exist in the background, the KAN module enhances the local sensitivity of the model while maintaining global constraints, making the contours of abnormal targets clearer. Finally, by integrating the KAN-based multipath collaborative module into $C5$, our MiKA-HAD framework is refined. By introducing RBF to enhance the model's nonlinear modeling capabilities, the model produces a cleaner background while retaining anomaly information, achieving optimal performance. Most notably, these results reveal synergistic interactions between our innovations rather than merely additive effects. The progressive performance gains demonstrate that our sample balancing strategy creates optimized conditions for the hybrid network to learn discriminative features, while the KAN-based module effectively exploits this enriched feature space through complementary modeling pathways. This confirms our architectural design philosophy of addressing both data distribution challenges and feature extraction limitations in a unified framework.

D. Parameter Analysis

In this study, we conducted a detailed sensitivity analysis on the key hyperparameters involved in our MiKA-HAD across all datasets. First, we investigated the impact of the min sample and min cluster size parameters in the HDBSCAN density clustering algorithm on network performance. As shown in Fig. 8, when the min sample and min cluster size are set to 2 and 5, respectively, the clustering algorithm optimally separates the majority and minority background samples, thus providing a clear distinction boundary for subsequent anomaly detection. This parameter combination effectively balances the heterogeneous distribution characteristics of background samples: the smaller minimum cluster size preserves more spectrally discrete minority background samples, while the moderate minimum cluster size prevents over-segmentation caused by excessively low-density thresholds. Furthermore, we analyzed the critical parameters in the process of minority sample augmentation—augmentation multiple and augmentation ratio. As depicted in Fig. 8, the network's detection performance is optimal when the augmentation multiple is 8 and the augmentation ratio is 10%. This finding reveals a critical insight: increasing the absolute number of minority samples to a specific threshold creates sufficient feature space density to influence model training, effectively shifting decision boundaries to better differentiate between minority backgrounds and anomalies. However, excessive augmentation beyond this optimal point provides diminishing returns and may introduce noise that degrades performance.

In addition, changes in the augmentation multiple have a more significant impact on the results than the other three hyperparameters, which is consistent with our expectations: changes in the other three hyperparameters lead to insufficient feature learning for certain samples, such as those not correctly clustered by HDBSCAN or not augmented by the sample augmentation method, while changes in the augmentation multiple directly affect the proportion of different types of background samples. This impact no longer pertains to only certain categories of background samples but affects all background samples. Therefore, maintaining a balanced quantity of different types of background samples is more important. This conclusion further validates our initial hypothesis that rebalancing the model's feature space can effectively improve detection performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:55:13 UTC from IEEE Xplore. Restrictions apply.

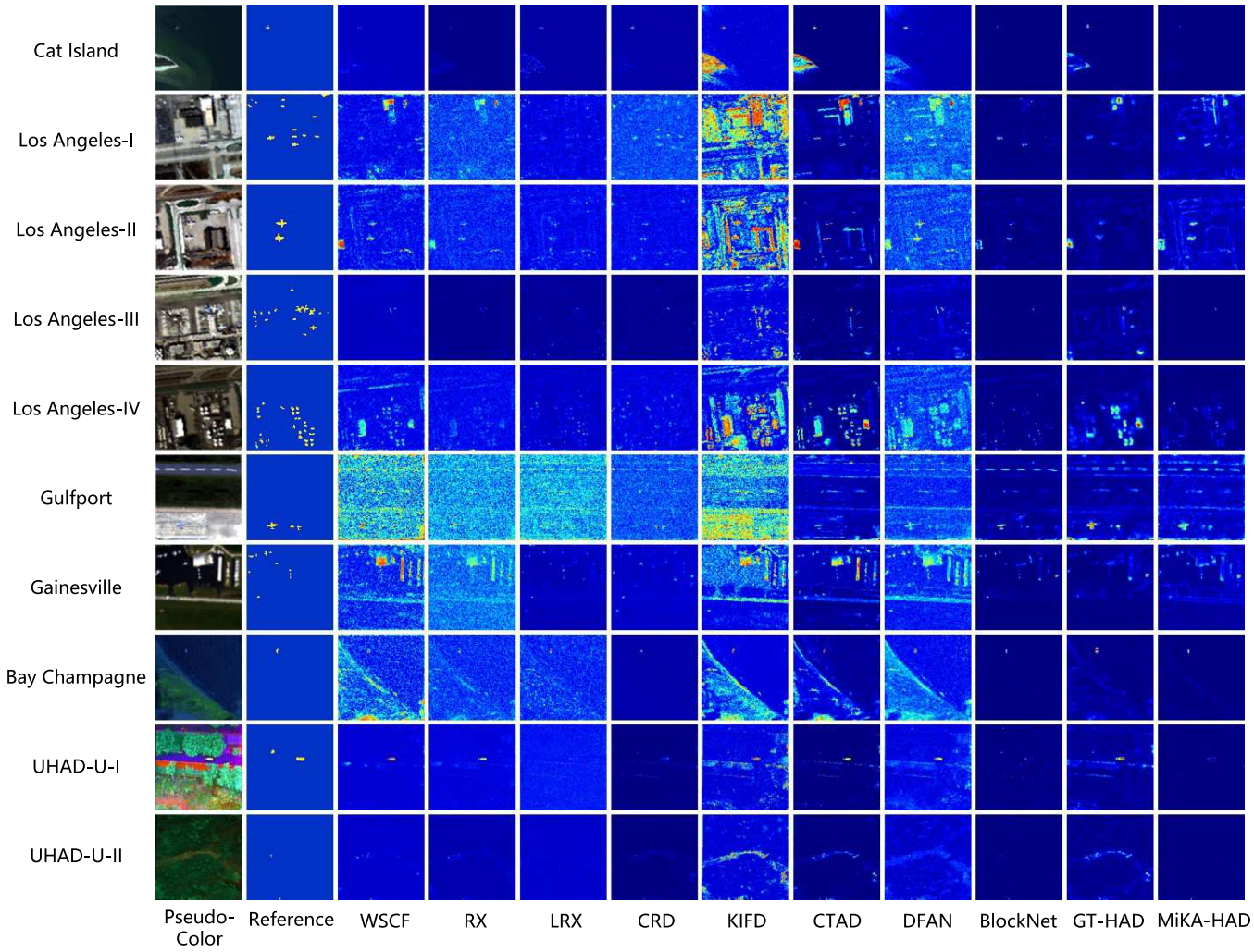


Fig. 9. Anomaly detection results of different methods on all datasets of $N(0, \text{Mean} \times 0.1)$ noise level.

E. Robustness to Noise

To validate MiKA-HAD's practical utility under real-world conditions, we conducted comprehensive anti-interference experiments by introducing controlled noise perturbations across all test datasets. Considering the diverse scene characteristics across different datasets, we design two levels of adaptive noise: Gaussian distributions $N(0, \text{Mean} \times 0.01)$ at approximately 60 dB and $N(0, \text{Mean} \times 0.1)$ at approximately 40 dB, where Mean represents the average pixel value of each dataset. Compared to fixed-decibel noise, Gaussian noise following preset distributions better emulates real sensor noise characteristics, providing more realistic robustness validation.

Table IV summarizes the AUC comparisons under two noise levels. Experimental results demonstrate that our method's exceptional robustness maintains superior performance across most test scenarios, even under severe noise conditions. Most notably, as noise intensity escalates from moderate $N(0, \text{Mean} \times 0.01)$ to severe $N(0, \text{Mean} \times 0.1)$, the performance gap between MiKA-HAD and competing methods widens significantly from an average AUC advantage of 0.0261–0.0293. This progressive enhancement of relative performance under increasing interference confirms our method's relative advantage in noisy environments.

Fig. 9 visualizes detection results under $N(0, \text{Mean} \times 0.1)$ noise. Taking the Gulfport dataset as an example, traditional statistical or collaborative methods fail to effectively identify anomalies, with only CTAD demonstrating partial noise suppression. Kernel-based KIFD exhibits excessive false alarms, limiting practical utility. Deep learning approaches DFAN suffer severe performance degradation under noise, while BlockNet and GT-HAD show incomplete anomaly annotations or localized false markings. In contrast, our method achieves effective noise suppression while maintaining complete anomaly annotation capability as a direct benefit of our minority-augmented sample strategy, preventing noise-induced false alarms and the KAN-based multipath architecture preserves discriminative feature extraction under noise perturbation. Specifically, the minority-augmented sampling strategy enhances discrimination between minority-class and anomalous samples. Within the KAN-based multipath architecture, B-splines ensure local feature precision while RBFs enable comprehensive extraction of high-dimensional nonlinear features, complemented by linear transformations that enforce

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:55:13 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
AUC FOR DIFFERENT METHODS ON ALL DATASETS OF DIFFERENT NOISE LEVELS. THE BEST PERFORMERS PER ROW ARE BOLDED; SECOND-BEST PERFORMERS ARE UNDERLINED

Dataset	WSCF [44]	RX [8]	LRX [45]	CRD [14]	KIFD [11]	CTAD [46]	DFAN [25]	BlockNet [47]	GT-HAD [48]	MiKA-HAD
Noise Level	$N(0, Mean \times 0.01)$									
Cat Island	0.9836	0.9749	0.9900	0.9697	0.9768	0.9286	0.9892	0.9698	0.9836	**0.9931**
Los Angeles-I	0.8606	0.8432	0.9120	**0.9549**	0.8751	0.7930	0.9056	0.9089	0.8569	**0.9549**
Los Angeles-II	0.9031	0.8732	0.8905	0.8451	0.9691	0.8814	0.9369	0.9464	0.8663	**0.9783**
Los Angeles-III	0.9232	0.9170	0.9084	0.9278	**0.9536**	0.8840	0.9318	0.9221	0.8610	0.9515
Los Angeles-IV	0.9532	0.9534	0.9236	0.9103	0.9678	0.9558	0.9334	0.8914	0.9324	**0.9803**
Bay Champagne	0.9988	0.9998	0.9981	0.9991	0.9985	0.9527	0.9998	0.9984	0.9833	**0.9999**
Gulfport	0.9337	0.9246	0.8126	0.8044	0.9811	0.9448	0.9887	**0.9961**	0.9470	0.9941
Gainesville	0.9471	0.9636	0.9532	0.9780	0.9488	0.9228	0.9713	0.9806	0.9353	**0.9834**
UHAD-U-I	0.8913	0.9091	0.8184	0.9463	0.9149	0.8420	0.8744	0.8862	0.9700	**0.9766**
UHAD-U-II	0.9530	0.9677	0.9335	0.9712	0.9622	0.8956	0.9648	0.9667	0.5423	**0.9961**
Average	0.9347	0.9326	0.9142	0.9306	0.9547	0.9000	0.9495	0.9466	0.8878	**0.9808**
Noise Level	$N(0, Mean \times 0.1)$									
Cat Island	0.9431	0.9445	0.9572	0.9519	0.9804	0.9643	0.9914	0.9653	0.9836	**0.9923**
Los Angeles-I	0.6813	0.6337	0.8625	0.8870	0.6788	0.6949	0.8386	0.9146	0.9003	**0.9403**
Los Angeles-II	0.8153	0.7461	0.8302	0.7396	0.8032	0.7468	0.8863	0.8744	0.9305	**0.9421**
Los Angeles-III	0.8644	0.8307	0.8675	0.8504	0.9220	0.8223	0.9203	0.9415	0.8588	**0.9499**
Los Angeles-IV	0.8709	0.8310	0.9051	0.8044	0.9363	0.9482	0.9240	0.8838	0.9425	**0.9850**
Gulfport	0.7664	0.7368	0.7192	0.7647	0.7092	0.8578	0.9497	0.9734	0.9660	**0.9795**
Gainesville	0.8608	0.8364	0.8988	0.9138	0.8069	0.9417	0.9659	**0.9884**	0.9326	0.9707
Bay Champagne	0.9499	0.9309	0.8911	0.9662	0.9552	0.9465	0.9865	0.9982	0.9932	**0.9999**
UHAD-U-I	0.7804	0.7886	0.6490	0.9481	0.8193	0.8034	0.7963	0.9118	0.9412	**0.9796**
UHAD-U-II	0.9412	0.8837	0.7920	0.9634	0.9125	0.9001	0.9401	0.9561	0.9928	**0.9948**
Average	0.8473	0.8162	0.8372	0.8789	0.8523	0.8626	0.9199	0.9407	0.9441	**0.9734**

fundamental global constraints—collectively ensuring robustness under complex noise interference.

V. CONCLUSION

This article proposes MiKA-HAD, a minority-augmented hybrid self-supervised framework based on KAN for HAD. By identifying minority-class background samples through density-based clustering and implementing augmentation strategies, our method explicitly guides feature learning to address the challenge of distinguishing minority-class backgrounds from anomalous targets in heterogeneous scenes. To tackle the challenges posed by high-dimensional data complexity and nonlinear feature extraction difficulties, we integrate AE, GAN, and a KAN-based multipath collaborative module to establish a self-supervised model, thereby enhancing the network's adaptability and robustness when processing complex high-dimensional data. Experimental results demonstrate that despite limitations in computational efficiency and scalability, our method achieves marked effectiveness across multiple datasets, demonstrating enhanced performance and robustness under various noise conditions. In future work, we will focus on optimizing resource consumption and enhancing scalability. We anticipate that this research will provide valuable insights for future explorations in HAD and related fields.

REFERENCES

[1] Y. Xu, L. Zhang, B. Du, and L. Zhang, "Hyperspectral anomaly detection based on machine learning: An overview," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 3351–3364, 2022.

[2] D. Ma, S. Xu, Z. Jiang, and Y. Yuan, "Central pixel-based dual-branch network for hyperspectral image classification," *Remote Sens.*, vol. 17, no. 7, p. 1255, Apr. 2025.

[3] Y. Yuan, C. Wang, and Z. Jiang, "Proxy-based deep learning framework for spectral–spatial hyperspectral image classification: Efficient and robust," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501115.

[4] C. McCann, K. S. Repasky, R. Lawrence, and S. Powell, "Multi-temporal mesoscale hyperspectral data of mixed agricultural and grassland regions for anomaly detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 131, pp. 121–133, Sep. 2017.

[5] D. Ma, Z. Liu, and Z. Jiang, "Variation autoencoder of spatial–spectral joint mask for hyperspectral anomaly detection," *IEEE Signal Process. Lett.*, vol. 32, pp. 1535–1539, 2025.

[6] M. T. Eismann, A. D. Stocker, and N. M. Nasrabadi, "Automated hyperspectral cueing for civilian search and rescue," *Proc. IEEE*, vol. 97, no. 6, pp. 1031–1055, Jun. 2009.

[7] X. Briottet et al., "Military applications of hyperspectral imagery," *Proc. SPIE*, vol. 6239, pp. 82–89, May 2006.

[8] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

[9] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[10] R. Zhao, B. Du, and L. Zhang, "A robust nonlinear hyperspectral anomaly detection approach," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 4, pp. 1227–1234, Apr. 2014.

[11] S. Li, K. Zhang, P. Duan, and X. Kang, "Hyperspectral anomaly detection with kernel isolation forest," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 319–329, Jan. 2020.

[12] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.

[13] R. Zhao, B. Du, and L. Zhang, "Hyperspectral anomaly detection via a sparsity score estimation framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 6, pp. 3208–3222, Jun. 2017.

[14] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:55:13 UTC from IEEE Xplore. Restrictions apply.

[15] W. Sun, C. Liu, J. Li, Y. M. Lai, and W. Li, "Low-rank and sparse matrix decomposition-based anomaly detection for hyperspectral imagery," *J. Appl. Remote Sens.*, vol. 8, no. 1, May 2014, Art. no. 083641.

[16] Y. Yang, S. Song, D. Liu, J. C.-W. Chan, J. Li, and J. Zhang, "Hyperspectral anomaly detection through sparse representation with tensor decomposition-based dictionary construction and adaptive weighting," *IEEE Access*, vol. 8, pp. 72121–72137, 2020.

[17] X. Ma, X. Zhang, N. Huyan, X. Tang, B. Hou, and L. Jiao, "Hyper-Laplacian regularized low-rank tensor decomposition for hyperspectral anomaly detection," in *Proc. IGARSS-IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2018, pp. 6380–6383.

[18] X. Hu et al., "Hyperspectral anomaly detection using deep learning: A review," *Remote Sens.*, vol. 14, no. 9, p. 1973, Apr. 2022.

[19] S. Wang, X. Wang, L. Zhang, and Y. Zhong, "Auto-AD: Autonomous hyperspectral anomaly detection network based on fully convolutional autoencoder," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5503314.

[20] M. E. Aghili, H. Ghassemian, and M. I. Arani, "Spectral–spatial anomaly detection in hyperspectral imagery based on dual-domain autoencoders," in *Proc. 13th Iranian/3rd Int. Mach. Vis. Image Process. Conf. (MVIP)*, Mar. 2024, pp. 1–5.

[21] H. Wen, X. Hu, and P. Zhong, "DSRNet: Diagonal subsampling reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5628916.

[22] Z. Li, Y. Wang, C. Xiao, Q. Ling, Z. Lin, and W. An, "You only train once: Learning a general anomaly enhancement network with random masks for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5506718.

[23] J. Zhong, W. Xie, Y. Li, J. Lei, and Q. Du, "Characterization of background-anomaly separability with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6017–6028, Jul. 2021.

[24] S. Xiao, T. Zhang, Z. Xu, J. Qu, S. Hou, and W. Dong, "Anomaly detection of hyperspectral images based on transformer with spatial–spectral dual-window mask," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1414–1426, 2023.

[25] X. Cheng et al., "Deep feature aggregation network for hyperspectral anomaly detection," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–16, 2024.

[26] Q. Guo, Y. Cen, L. Zhang, Y. Zhang, and Y. Huang, "Hyperspectral anomaly detection based on spatial–spectral cross-guided mask autoencoder," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9876–9889, 2024.

[27] J. Li, X. Wang, S. Wang, H. Zhao, and Y. Zhong, "One-step detection paradigm for hyperspectral anomaly detection via spectral deviation relationship learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5517515.

[28] J. Wang, S. Guo, Z. Hua, R. Huang, J. Hu, and M. Gong, "CL-CaGAN: Capsule differential adversarial continual learning for cross-domain hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5517315.

[29] W. Chen, X. Zhi, S. Jiang, Y. Huang, Q. Han, and W. Zhang, "DWSDiff: Dual-window spectral diffusion for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504617.

[30] X. He et al., "Global-to-local spatial–spectral awareness transformer network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5530422.

[31] X. Fu, T. Zhang, J. Cheng, and S. Jia, "MMR-HAD: Multiscale mamba reconstruction network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5516914.

[32] S. Liu, L. Peng, X. Chang, Z. Wang, G. Wen, and C. Zhu, "Adaptive dual-domain learning for hyperspectral anomaly detection with state-space models," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5503719.

[33] K. Jiang, W. Xie, Y. Li, J. Lei, G. He, and Q. Du, "Semisupervised spectral learning with generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 5224–5236, Jul. 2020.

[34] Z. Wu et al., "Background-guided deformable convolutional autoencoder for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531816.

[35] S. Arisoy, N. M. Nasrabadi, and K. Kayabol, "Unsupervised pixel-wise hyperspectral anomaly detection via autoencoding adversarial networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[36] T. Jiang, Y. Li, W. Xie, and Q. Du, "Discriminative reconstruction constrained generative adversarial network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4666–4679, Jul. 2020.

[37] F. Girosi and T. Poggio, "Representation properties of networks: Kolmogorov's theorem is irrelevant," *Neural Comput.*, vol. 1, no. 4, pp. 465–469, Dec. 1989.

[38] Z. Liu et al., "KAN: Kolmogorov–Arnold networks," 2024, *arXiv:2404.19756*.

[39] N. Raza Shah et al., "Hyperspectral anomaly detection: A performance comparison of existing techniques," *Int. J. Digit. Earth*, vol. 15, no. 1, pp. 2078–2125, Dec. 2022.

[40] L. McInnes, J. Healy, and S. Astels, "Hdbscan: Hierarchical density based clustering," *J. Open Source Softw.*, vol. 2, no. 11, p. 205, Mar. 2017.

[41] Y. Yuan, Y. Zhao, and D. Ma, "NACAD: A noise-adaptive context-aware detector for remote sensing small objects," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 1001413.

[42] Y. Altmann, N. Dobigeon, J-Y. Tourneret, and S. McLaughlin, "Nonlinear unmixing of hyperspectral images using radial basis functions and orthogonal least squares," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2011, pp. 1151–1154.

[43] H. Goldberg, H. Kwon, and N. M. Nasrabadi, "Kernel eigenspace separation transform for subspace anomaly detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 4, pp. 581–585, Oct. 2007.

[44] F. Verdoja and M. Grangetto, "Graph Laplacian for image anomaly detection," *Mach. Vis. Appl.*, vol. 31, nos. 1–2, p. 11, Feb. 2020.

[45] J. M. Molero, E. M. Garzon, I. Garcia, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[46] L. Gao, X. Sun, X. Sun, L. Zhuang, Q. Du, and B. Zhang, "Hyperspectral anomaly detection based on chessboard topology," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505016.

[47] D. Wang, L. Zhuang, L. Gao, X. Sun, M. Huang, and A. Plaza, "BockNet: Blind-block reconstruction network with a guard window for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531916.

[48] J. Lian, L. Wang, H. Sun, and H. Huang, "GT-HAD: Gated transformer for hyperspectral anomaly detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 2, pp. 3631–3645, Feb. 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 12:55:13 UTC from IEEE Xplore. Restrictions apply.